

Agent Role Taxonomy in Software Engineering

Derived from 19 SOTA Multi-Agent Systems

1. Taxonomy Overview

The following table summarizes the hierarchical organization of agent roles across 19 state-of-the-art multi-agent frameworks in software engineering.

Layer	Purpose	Representative Roles	Example Systems
Orchestration	Manage coordination and task flow	Manager, Supervisor, Project Coordinator	MetaGPT, Magentic-One, CodeAgent
Analysis & Planning	Define requirements, goals, and decomposition	Product Manager, Planner, Requirement Analyst	MetaGPT, MapCoder
Design	Specify architectures and high-level structures	Architect, Designer, Modeler	MetaGPT, PotPie
Implementation	Create executable artifacts	Coder, Implementer, World-Model Coder	WorldCoder, Tree-of-Code
Quality Assurance	Evaluate and verify outputs	Reviewer, Tester, Debugger	CodeAgent, PotPie
Knowledge & Documentation	Maintain repository and documentation	Knowledge Curator, Writer, Explainer	PotPie, MetaGPT
Operations & Tooling	Execute, monitor, interact with environments	Executor, Environment Interface Agent	Tree-of-Code, WorldCoder

2. Functional Dimensions

Each agent role can be analyzed along three fundamental dimensions: initiative, collaboration, and artifact focus.

Dimension	Description	Role Examples
Initiative	Autonomous decision-making level (proactive vs reactive)	Planner (proactive), Reviewer (reactive)
Collaboration Mode	Interaction pattern among agents	Manager/Worker (hierarchical), Critic/Collaborator (peer)
Artifact Focus	Primary artifact or context handled by the agent	Architect (design), Coder (code), Curator (knowledge)

3. Mapping Explicit and Inferred Roles

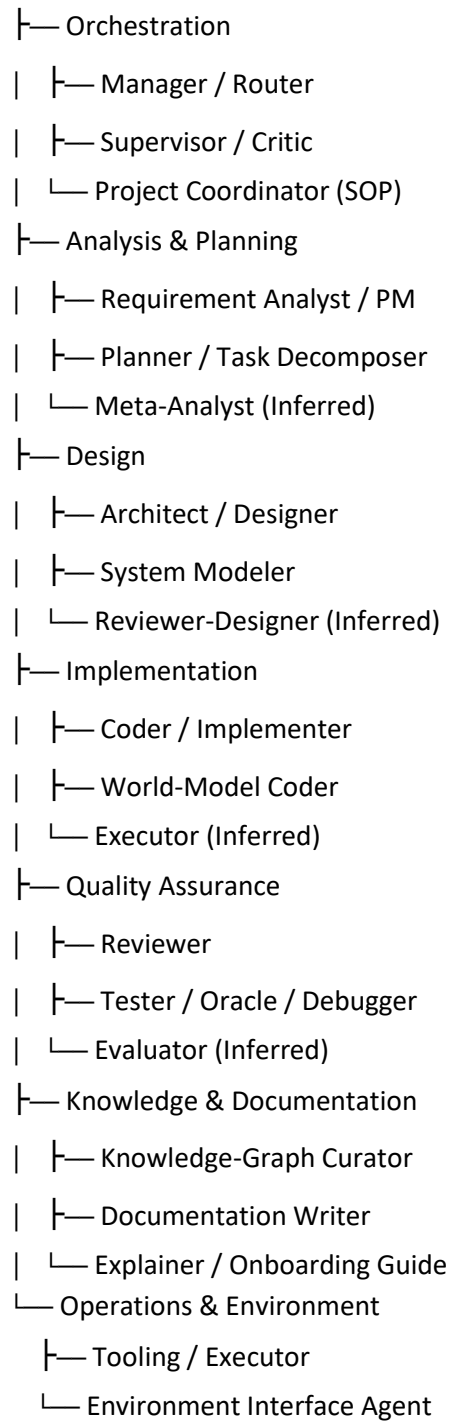
Some systems explicitly define agent roles, while others require inferred categorization based on functionality and workflow context.

Paper Type	Explicit / Inferred Roles	Rationale
Multi-Agent Systems	Explicit: Manager, Planner, Coder, Reviewer	Defined within architecture
Conceptual / Taxonomy Papers	Inferred: Meta-Analyst, Pattern Cataloguer	Conceptual frameworks only
Empirical / Productivity Studies	Inferred: Observer, Evaluator	Observes developer-agent dynamics
Surveys / Reviews	Inferred: Synthesizer, Classifier	Synthesizes previous literature
Testing Frameworks	Explicit: Generator, Executor, Oracle	Roles defined for validation
Knowledge-Grounded Systems	Explicit: Debugger, Tester, Curator	Repository-aware specialization

4. Core Taxonomy Tree

The tree below illustrates the hierarchical relationships among agent role categories.

Agent Roles



5. Insights and Applications

The taxonomy reveals that modern agent ecosystems in software engineering increasingly adopt hybrid roles. High-frequency roles such as Planner, Coder, and Reviewer dominate practical frameworks, while new roles such as Knowledge-Graph Curator and World-Model Coder are emerging in repository-aware systems (e.g., PotPie, WorldCoder).

Future agent-based architectures may employ adaptive hierarchies, allowing agents to shift dynamically between planning, execution, and quality assurance roles depending on context and workload.

6. Paper-to-Role Mapping Matrix

This section maps each paper (A1–C18 + PotPie) to the agent roles it explicitly or implicitly defines, along with the corresponding taxonomy layers.

Paper (ID + Title)	Primary Roles	Taxonomy Categories
A1 – MetaGPT	Product Manager, Architect, QA Engineer	Orchestration, Planning, Design, QA, Knowledge
A2 – Magentic-One	Manager, Specialist Agents	Orchestration, Operations
A3 – Self-Organized Agents	(Inferred) Self-Coordinator, Optimizer	Orchestration, Implementation
A4 – MapCoder	Planner, Coder, Debugger	Planning, Implementation, QA
A5 – Händler (Taxonomy Paper)	(Inferred) Meta-Analyst, Framework Designer	Knowledge, Planning
B6 – CodeAgent	Supervisor, Reviewer	Orchestration, QA
B7 – Multi-Agent LLM SE Framework	(Inferred) Coordinator, Evaluator	Orchestration, QA
B8 – WorldCoder	Planner+Coder	Implementation, Operations
B9 – Tree-of-Code	Planner/Executor	Planning, Implementation
B10 – LLM Agents Survey	(Inferred) Synthesizer	Knowledge
C11 – MAS for SE Review	(Inferred) Synthesizer, Classifier	Knowledge, QA
C12 – AI Agents Architecture Survey	(Inferred) Framework Analyst	Knowledge, Design
C13 – METR Productivity Study	(Inferred) Observer, Evaluator	QA, Knowledge
C14 – Apiiro AI Risk Report	(Inferred) Evaluator, Risk Assessor	QA, Operations
C15 – Agentless Discussion	(Inferred) Analyst, Critic	Knowledge, QA
C16 – Design Space of AI Coding Assistants	(Inferred) UX Analyst	Design, Planning

C17 – Autonomous Agents for Testing	Generator, Executor, Oracle	QA, Operations
C18 – User Perception Study	(Inferred) Observer, Analyst	Knowledge, QA
P – PotPie	Debugger, Tester, Designer, KG Curator	Design, QA, Knowledge, Operations

7. Role Coverage by Layer

The table below summarizes how many papers contribute to each taxonomy layer and lists their corresponding IDs.

Taxonomy Layer	Paper IDs	Count
Orchestration	A1, A2, A3, B6, B7	5
Analysis & Planning	A1, A4, A5, B9, C16	5
Design	A1, C12, C16, P	4
Implementation	A1, A3, A4, B8, B9	5
Quality Assurance	A1, A4, B6, C13, C14, C15, C17, C18, P	9
Knowledge & Documentation	A1, A5, B10, C11, C12, C15, C18, P	8
Operations & Tooling	A2, B8, B9, C14, C17, P	6

This mapping highlights the dominance of QA- and Knowledge-oriented agents, showing a strong trend toward autonomous evaluation, documentation, and explainability within multi-agent SE frameworks.